Unit – 3
Mixed Reality
Optical See-Through Displays in MR

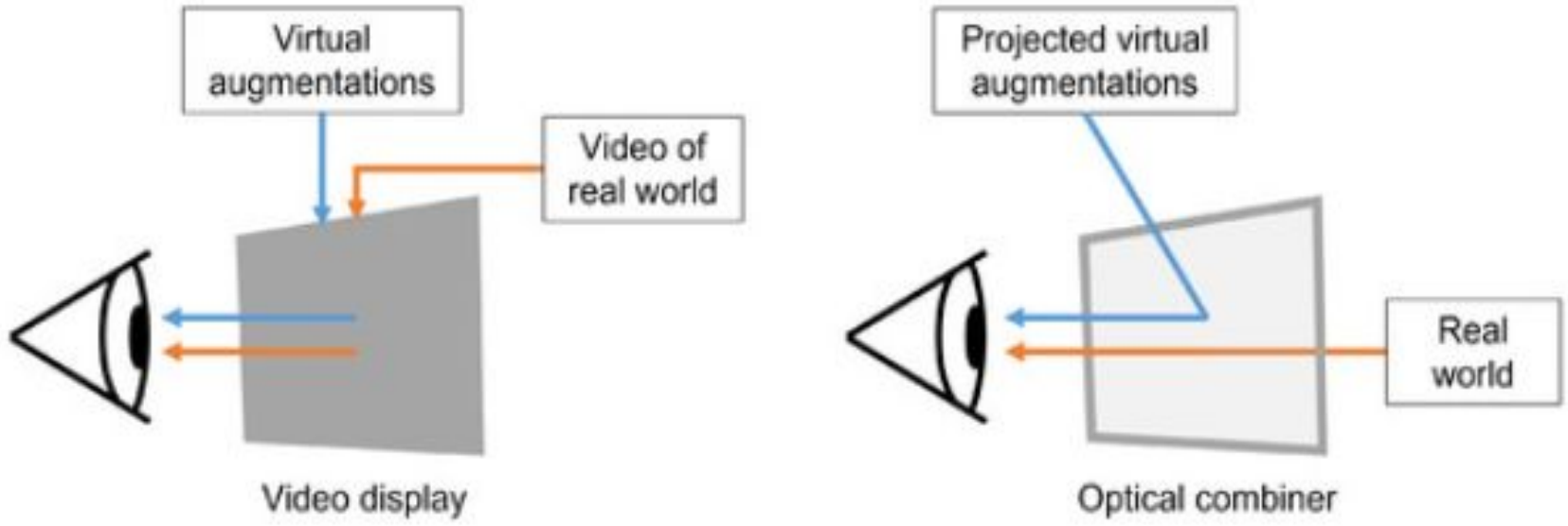
Optical See-Through Displays (OST Displays)

- Optical See-Through Displays are MR display systems in which the user views the real world through a transparent display while computer-generated virtual objects are superimposed onto the real environment.

OST invention:

- Optical see-through HMDs were developed and first introduced in the 1960s, with Ivan Sutherland the father of CG

- He demonstrated the first computer graphics-based HMD in 1968 using miniaturized cathode-ray tubes as stereoscopic displays and a mechanical tracker to provide head position and orientation in real-time.



(a) Video see-through display

(b) Optical see-through display

Working of OST Display:

- User looks through **transparent glasses or visor**
- **Sensors and cameras** track the environment, Detect position, objects, and user movement.
- MR Software determines where virtual objects should appear.
- A micro-display or projector creates digital graphics.
- Optical combiner merges images as virtual light is reflected into the user's eyes.
- User sees **real world + virtual objects together**

Working of OST Display:

- **1. Real World Viewing**
  - The user looks through a transparent optical lens.
  - Real-world light passes directly into the eyes without any digital processing.
  - This gives a natural, real-time view of the environment.
- **2. Virtual Content Generation**
  - A micro-display (like OLED/LCD) generates digital images.
  - These images include:
    - 3D objects
    - Text and graphics
    - Interactive elements

Working of OST Display:

- **3. Image Projection**
  - The generated image is projected into the optical system.
  - This is done using:
    - Mini projectors
    - Light engines
- **4. Optical Combination (Key Step)**
  - A beam splitter / waveguide / combiner merges:
    - Real-world light
    - Virtual image light
  - This component partially reflects virtual light and transmits real-world light.

Working of OST Display:

- **5. Alignment & Registration**
  - Sensors (camera, gyroscope, accelerometer) track:
    - Head movement
    - Eye position
    - Surroundings
  - The system adjusts the virtual image so it aligns correctly with real objects.

- **6. Rendering & Processing**
  - A processor continuously:
    - Updates virtual content
    - Adjusts perspective
    - Maintains real-time interaction

Working of OST Display:

- **7. Final Perception**
  - The brain combines:
    - Real-world view (direct)
    - Virtual overlay (projected)
  - User experiences a merged Mixed Reality environment.

- **Working Flow (Simple)**

Real Scene →

Transparent Lens →

Virtual Image Generation →

Optical Combiner →

Eye →

Brain Perception

Real Scene

Transparent Lens

Virtual Image Generation

Optical Combiner

Eye

Brain Perception

Components of OST Displays

- Transparent Lens
  - Allows the direct view of real world.
- Micro Display / Projector
  - Generates virtual images.
- Optical Combiner (Beam Splitter / Waveguide)
  - Merges real and virtual images
- Sensors and Cameras
  - Track environment and user movement.
- Processing Unit
  - Controls rendering and alignment of 3D virtual objects.

Thank You…!!!